HOSSEIN SHIRALI

Doctoral Researcher

Karlsruhe, Germany

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Google Scholar | GitHub | LinkedIn

RESEARCH INTERESTS

Computer vision and deep learning for biodiversity research, with a focus on automated insect digitization, species identification, and estimation of morphological traits and biomass. I develop AI workflows and integrate them into robotic imaging systems for ecological research and forest health surveillance.

CURRENT POSITION

Karlsruhe Institute of Technology (KIT) — Karlsruhe, Germany
Doctoral Researcher — *Mar. 2023 - Present*

EDUCATION

Karlsruhe Institute of Technology (KIT) — Karlsruhe, Germany
Ph.D. Candidate — *Present*
Research: biodiversity research of invertebrates using deep learning methods.

- Supervised by apl. Prof. Dr. Christian Pylatiuk and apl. Prof. Dr. Markus Reischl.

University of Bologna — Bologna, Italy
Master of Electronic Technologies for Big Data and Internet of Things
Grade: 110/110. — *Oct. 2021*
Fields: statistics and architectures for big data processing and communications; signal acquisition and processing.

Shahid Chamran University of Ahvaz — Iran
Bachelor of Electronic Engineering — *2017*
Fields: computer architecture, digital systems, and electronics.

RESEARCH PROJECTS

Entomoscope and DiversityScanner

- Lead AI development for the Entomoscope photomicroscope and contribute to the DiversityScanner robotic ecosystem for high-throughput insect digitization and analysis.
- Design end-to-end deep learning pipelines for species identification, sex classification, orientation detection, and anatomical segmentation of morphologically challenging dark taxa.
- Collaborate with engineers, taxonomists, and ecologists to integrate AI workflows into robotic devices and transition systems from prototype to deployment.

InsectMorphoAI

- Develop an image-based software suite to estimate insect length, volume, and biomass from 2D images for large-scale ecological assessment.

FORSAID

- Contribute to the EU Horizon Europe project Forest Surveillance with Artificial Intelligence and Digital Technologies (Grant No. 101134200).
- Build deep learning classifiers for regulated forest pest species identification from trap images within WP3, "Digital technologies for ground detection and surveillance of regulated pests."

Publications

Peer-Reviewed Journal Articles

[J1] Ascenzi, A., **Shirali, H.**, Di Lorenzo, N., Nania, D., Wührl, L., Pylatiuk, C., Meier, R., & Cerretti, P. (2026). An AI-driven biomass estimation tool enabling population- and specimen-level variation analyses: A case study on parasitoid flies. *Entomologia Experimentalis et Applicata*, Published online 8 September 2026.
doi: 10.1111/eea.70190

[J2] Saur, L., von Pawlowski, M., Gengenbach, U., Sieber, I., **Shirali, H.**, Wührl, L., Weng, X., Kiko, R., & Pylatiuk, C. (2026). Classification of microplastic particles in water using polarized light scattering and machine learning methods. *Journal of Hazardous Materials Advances*, 23, 101343.
doi: 10.1016/j.hazadv.2026.101343

[J3] Klug, N., Kramer, M., Mazrek, F., Wührl, L., **Shirali, H.**, Meier, R., & Pylatiuk, C. (2026). Automated photogrammetric close-range imaging of small invertebrates using acoustic levitation. *Automatisierungstechnik*, 74(7), 564-576.
doi: 10.1515/auto-2025-0139

[J4] **Shirali, H.**, Ascenzi, A., Wührl, L., Beyer, N., Di Lorenzo, N., Vaccarella, E., Klug, N., Meier, R., Cerretti, P., & Pylatiuk, C. (2026). InsectMorphoAI: A deep learning framework for automated insect morphometrics and biomass estimation with taxon-specific volumetric validation. *Ecological Informatics*, 96, 103854.
doi: 10.1016/j.ecoinf.2026.103854

[J5] Caruso, V., **Shirali, H.**, Bouget, C., Cerretti, P., Curletti, G., de Groot, M., Groznik, E., Gutowski, J. M., Pylatiuk, C., Plewa, R., Roques, A., Sallé, A., Sweeney, J., Van Rooyen, K., Wührl, L., & Rassati, D. (2026). Image-based recognition using advanced neural networks can aid surveillance of Agrilus jewel beetles. *NeoBiota*, 105, 319-336.
doi: 10.3897/neobiota.105.180959

[J6] **Shirali, H.**, Wührl, L., Lee, L., Klug, N., Meier, R., Pylatiuk, C. et al. (2026). Automated specimen triage for dark taxa: Deep learning enables orientation, sex identification and anatomical segmentation from robotic imaging. *Systematic Entomology*, 51(1), e70039.
doi: 10.1111/syen.70039

[J7] Wührl, L., Keller, L., Klug, N., **Shirali, H.**, Meier, R., & Pylatiuk, C. (2024). Automated handling of biological objects with a flexible gripper for biodiversity research. *Automatisierungstechnik*, 72(7), 672-678.
doi: 10.1515/auto-2023-0238

[J8] **Shirali, H.**, Hübner, J. J., Both, R., Raupach, M. J., Reischl, M., Schmidt, S., & Pylatiuk, C. (2024). Image-based recognition of parasitoid wasps using advanced neural networks. *Invertebrate Systematics*, 38(6), IS24011.
doi: 10.1071/IS24011

Preprints and Papers in Submission

[P1] **Shirali, H.**, Boese, N. E., Kramer, M., Wang, J., Klug, N., Mikut, R., Meier, R., & Pylatiuk, C. (2026). Entomoscope 2.0 and ENIMAS 2.0: An open-source, AI-integrated platform for rapid and affordable insect digitization. *bioRxiv*.
doi: 10.64898/2026.01.23.701290

Industry Experience

Modis / Baker Hughes Italy, Remote
Data Scientist Consultant *Feb. 2022 - Mar. 2023*

- Developed an optimization algorithm to improve turbine maintenance efficiency.
- Led cloud data migration and implemented enhanced data security protocols.
- Built automation tools using Microsoft Power Platform and delivered internal O365 training.

Plasive Technologies Bologna, Italy
Junior Data Scientist *Feb. 2021 - Oct. 2021*

- Designed and trained CNN models for semantic segmentation of satellite imagery.
- Developed a carbon stock estimation method using remote sensing and data analytics, and constructed a multidimensional dataset for satellite image research.

Kiwitron Bologna, Italy
Machine Learning Intern *Sep. 2020 - Jan. 2021*

- Developed a semi-automatic image labeling method using point cloud data for my master's thesis; achieved state-of-the-art performance in image annotation tasks.

Conferences and Presentations

Entomology Congress (DGaaE) 2025
Geisenheim, Germany

- Presentation: “AI-driven advances in Species Identification and Biomass Analysis”

27th International Congress of Entomology (ICE) 2024
Kyoto, Japan

- Presentation: “AI as a Catalyst in Entomological Research by Simplifying Species Identification”

Helmholtz Artificial Intelligence Conference 2024
Düsseldorf, Germany

- Poster: “Advancing Biodiversity Research with AI-Driven Automation”

Helmholtz Imaging Conference 2023
Hamburg, Germany

- Poster: “Automated Biodiversity Research”

Skills and Languages

Machine learning and vision: Python, scikit-learn, TensorFlow, Keras, PyTorch, ONNX, OpenCV, YOLO

Infrastructure and tools: Docker, Kubernetes, Git, CI/CD, AWS, HPC/SLURM, Streamlit, Gradio, VS Code, Jupyter

Expertise: Deep learning, computer vision, model deployment, data and image analysis, neural network design, scientific computing, robotic imaging integration

Languages: Persian (native), English (B2), German (A2)

References

Prof. Dr. Christian Pylatiuk
Group Leader Biomedical Engineering & Robotics
Institute for Automation and Applied Informatics (IAI)
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Prof. Dr. Rudolf Meier
Head of the Center for Integrative Biodiversity Discovery
Museum für Naturkunde Berlin
Professor at Humboldt University zu Berlin
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